Project Description

Project Name: PegGuard

An Autonomous, Stateful-Watermark, Regime-Switching & Recovery-Risk-Adjusted Quantitative Arbitrage Framework Across Ethereum, Arbitrum, and FISCO BCOS

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Summary

Overview: Autonomous Multi-Chain Peg Arbitrage & Capital Preservation

Modern financial markets feature rapid growth in tokenized Real-World Assets (RWAs) and pegged instruments across **Ethereum Mainnet (L1)** and **Arbitrum One (L2)**, spanning fiat-backed and decentralized stablecoins (USDC, USDT, DAI, FRAX, USDe, crvUSD, GHO, LUSD), liquid staking and restaking derivatives (stETH, wstETH, WBETH, rETH, cbETH, weETH, ezETH), wrapped Bitcoin (WBTC, cbBTC, tBTC), and tokenized commodities/treasuries (PAXG, XAUT, USDY).

Conventional quantitative strategies make a fatal conceptual error: **they treat statistical mean reversion as synonymous with riskless financial arbitrage**. Buying a discounted pegged asset is not pure arbitrage if the underlying peg mechanism itself is impaired, if carry and panic slippage erase the spread, or if the hedge leg introduces short-squeeze risk.

To resolve these flaws, **PegGuard** operates as a **User-Toggled Autonomous Arbitrage Daemon** equipped with **Stateful Unprofitable-Price Watermark Memory** and a **5-Layer Quantitative & Multi-Modal Risk Funnel**. When toggled ON, PegGuard continuously scans all target assets across Ethereum, Arbitrum, and centralized venues, routing approved trades through asset-specific perpetual hedges or an institutional **FISCO BCOS Par-Capped Repo Facility**.

Problem Statement: The Five Structural Failure Modes

1. **The Fair-Value vs. Market-Consensus Fallacy:** Relying purely on moving averages ignores sudden reserve impairment (e.g., bank lockups), while relying purely on theoretical par value ($1.00) ignores persistent market-priced risk discounts and accruing staking yields.
2. **Regime Blindness & Broken Mean-Reversion:** Single-state Ornstein-Uhlenbeck (O-U) models calibrate reversion speed (θ) on calm periods. When a peg enters fundamental collapse (e.g., Terra/UST), historical reversion speed becomes meaningless, leading bots to buy a terminal death spiral.
3. **Carry-Cost Bleed, Panic Slippage & Recovery Risk:** Apparent arbitrage spreads evaporate once accounting for multi-week borrow drag, redemption queue delays, L2 order-book depth contraction during panic, and expected Loss Given Default (LGD).
4. **Flawed Delta-Neutral Hedging & Stablecoin Contagion:** Shorting BTC/ETH to hedge a fiat stablecoin (USDC) introduces massive directional risk, whereas shorting a rival stablecoin (USDT) on retail perps exposes the trader to flight-to-safety short squeezes (>$1.04) and cross-stablecoin contagion.
5. **Stateless Re-Evaluation Spam on Shallow Dislocations:** Conventional bots repeatedly run expensive econometric and NLP checks every second on the exact same unprofitable de-peg price, exhausting compute and API limits instead of memorizing the unprofitable price floor and waiting for a deeper discount.

Autonomous Daemon & Stateful Watermark Memory

Rather than requiring manual CLI invocations for single asset pairs, **PegGuard** is engineered as a continuous, event-driven background daemon controlled by a live user state switch and an intelligent anti-spam price memory cache.

1. Live User Master Toggle (ArbitrageEngineEnabled ∈ {True, False})

The user controls a single master operational switch that can be flipped **ON** or **OFF** at any time without restarting the infrastructure:

- **When Enabled (ON):** The daemon continuously sweeps the entire multi-chain watchlist across Ethereum L1, Arbitrum L2, and Binance, evaluating real-time consensus price ratios $\mathbf{R_{cons}(t)}$ and standardized anomaly scores $\mathbf{Z(t)}$.
- **When Disabled (OFF):** The scanner immediately halts new trade entries while continuing to monitor open hedged positions until target peg convergence or manual unwind.

2. Stateful Unprofitable-Price Watermark Memory (WatermarkFloor_i)

When a monitored token de-pegs ($\mathbf{|Z_i(t)| \geq 2.0}$), PegGuard triggers Layers 2, 3, and 4 to verify whether the net trade is profitable after borrow APR, signed funding, taker fees, panic slippage, and default risk. If the trade is **not profitable** at the current de-pegged ratio $\mathbf{R_{rejected,i}}$ (e.g., USDC de-pegs to $0.9940, yielding -6 bps Net EV after fees and carry), re-evaluating the exact same trade every second is wasteful.

PegGuard solves this via a **Stateful Unprofitable-Price Watermark Cache**:

> If Rejected at $R_{rejected,i}$: $\text{WatermarkFloor}_i = \min[\, R_{rejected,i} - \Delta_{step} \,,\ R_{breakeven,i}(\text{Carry, Fees, LGD}) \,]$
> Re-Evaluation Trigger Rule : Run Layers 2–4 iff $|Z_i(t)| \geq 2.0$ AND $R_{cons,i}(t) < \text{WatermarkFloor}_i$

Daemon State / Condition	Mathematical Trigger	PegGuard Autonomous Action
Normal Peg Regime	$\mathbf{\lvert Z_i(t)\rvert < 1.5}$	Clears any cached watermark ($\texttt{WatermarkFloor}_i$ = None). Continues lightweight L1 scanning.
Fresh Dislocation	$\mathbf{\lvert Z_i(t)\rvert \geq 2.0}$ and $\texttt{WatermarkFloor}_i$ is None	Passes asset through Layers 2, 3 & 4 to evaluate duration, signed carry EV, and solvency.
Unprofitable Dislocation Noted	$\mathbf{EV_{rec} < \text{Hurdle}}$ at price $\mathbf{R_{eval}}$	Records $\texttt{WatermarkFloor}_i = R_{eval} - \Delta$. Suppresses repeat Layer 2–4 calls at or above this price.
Stagnant De-Peg (Skipped)	$\mathbf{\lvert Z_i(t)\rvert \geq 2.0}$ and $\mathbf{R_{cons,i}(t) \geq \text{WatermarkFloor}_i}$	**O(1) Instant Skip:** Price has not dropped lower than the noted unprofitable level; zero wasted compute.
Deeper Discount Re-Check	$\mathbf{\lvert Z_i(t)\rvert \geq 2.0}$ and $\mathbf{R_{cons,i}(t) < \text{WatermarkFloor}_i}$	**Wakes Up Pipeline:** Price dropped lower! Re-runs Layers 2–4 to check if the wider spread is now profitable.
Hard Solvency Veto Lockout	$\mathbf{S_t = 3}$ or $\mathbf{P(\text{Insolvency}) \geq 0.60}$	Sets $\texttt{WatermarkFloor}_i = 0.0$ (LOCKED). Blocks all buying regardless of how low the price falls.

Why Stateful Watermarking is Essential: During prolonged liquidity squeezes, a stablecoin or LST may hover at a shallow discount (e.g., 0.993) for hours before capitulating to 0.960. PegGuard evaluates 0.993 once, memorizes that 0.993 is unprofitable, and sleeps on that pair until the price drops below the break-even watermark.

Multi-Chain Asset Universe (Ethereum, Arbitrum & FISCO BCOS)

When enabled, PegGuard continuously monitors 22+ pegged, staked, and tokenized assets across **Ethereum Mainnet (L1)** and **Arbitrum One (L2)**, leveraging L1 for primary redemption/solvency telemetry, Arbitrum L2 for sub-second low-gas execution, and **FISCO BCOS** for institutional par-capped repo hedging and immutable risk auditing.

1. Class B1: Fiat-Backed & Decentralized Stablecoins (Hedged via FISCO BCOS Par-Capped Repo)

Pegged Asset	Real Anchor	Peg Mechanism & Backing	Ethereum (L1) & Arbitrum (L2) Venues	FISCO BCOS & PegGuard Role
USDC (Circle)	**USD** (T-Bills/Cash)	1:1 Fiat-Backed (Circle primary mint/burn)	**L1:** Curve 3pool, UniV3, Aave V3 **Arb:** Native USDC, Curve 2pool, Camelot	Primary Par-Repo Collateral; cross-chain state lock with FISCO BCOS vault.
USDT (Tether)	**USD** (T-Bills/Repo)	1:1 Fiat-Backed (0.10% redemption fee)	**L1:** Curve 3pool, UniV3, Aave V3 **Arb:** Curve 2pool, UniV3, Aave V3	Par-Capped Short Liability Benchmark (**min[P_{USDT}, 1.000]** prevents >$1.00 squeeze).
DAI / USDS (Sky)	**USD** (Crypto/RWA/PSM)	1:1 Soft-Pegged via USDC PSM & T-Bills	**L1:** Curve 3pool, Spark, Aave V3 **Arb:** Curve, UniV3, Balancer, Aave V3	Contagion Matrix (β_{ij}) tracked asset (high reserve correlation to USDC).
FRAX (Frax)	**USD** (Hybrid RWA/AMO)	1:1 Hybrid Collateralized (Curve AMO + sFRAX)	**L1:** Curve FRAX/USDC, Fraxlend **Arb:** Curve, Camelot, Ramses, Fraxlend	Contagion Matrix (β_{ij}) tracked asset (coupled via Curve FRAXBP).
USDe (Ethena)	**USD** (Spot + Short Perp)	1:1 Delta-Neutral Synthetic Dollar	**L1:** Curve USDe/USDC, Morpho, Pendle **Arb:** Curve, Camelot, Pendle, Morpho	High-carry basis target; Layer 3 compares Ethena funding vs. FISCO repo cost.
crvUSD & GHO	**USD** (Overcollateralized)	LLAMMA soft-liquidation (Curve) / Aave GSM	**L1 & Arb:** Curve, Balancer, UniV3, Aave V3	Ideal Layer 1 Dual-Anchor cases (structural discounts vs. par solvency).
LUSD, PYUSD, EURC	**USD / EUR**	100% ETH floor (LUSD) / Regulated Fiat (PYUSD, EURC)	**L1 & Arb:** Curve PayPool, LUSD/3Crv, UniV3, Aave V3	LUSD acts as inverse flight-to-safety indicator; PYUSD/EURC for multi-currency repo.

2. Class A1: Liquid Staking & Restaking Derivatives — LSTs & LRTs (Hedged via Short ETH-PERP)

Pegged Asset	Real Anchor	Peg Type & Fair-Value Growth	Ethereum (L1) & Arbitrum (L2) Venues	Layer 1 FV_{fund} & FISCO BCOS Role
stETH & wstETH (Lido)	**ETH** (Beacon Staked)	Rebasing 1:1 (stETH) / Yield-Accruing (**FV_{fund} ≈ 1.18+ ETH**)	**L1:** Curve stETH/ETH, UniV3, Aave V3 **Arb:** Deepest L2 LST (UniV3, Balancer, Camelot)	Primary Arbitrum LST target; tracks Lido withdrawal queue delay (τ_{redeem}).
WBETH (Binance)	**ETH** (Binance Staked)	Yield-Accruing Ratio (**FV_{fund} ≈ 1.085+ ETH**)	**L1 & CEX:** Binance Spot, Curve WBETH/ETH, UniV3	Dual-path triangulation benchmark; Oct 2025 flash-crash validation asset.
rETH & cbETH	**ETH** (RocketPool/Coinbase)	Yield-Accruing Ratios (**1.12+ / 1.07+ ETH**)	**L1 & Arb:** UniV3, Balancer, Curve, Aave V3	Layer 4 FinBERT monitors custodial/regulatory news vs. on-chain node burns.
weETH & ezETH (LRTs)	**ETH** (EigenLayer Restaked)	Yield-Accruing Restaking Ratios (**1.05+ ETH**)	**L1 & Arb:** Curve, UniV3, Camelot, Pendle, Aave V3	Models 7-day EigenLayer escrow lag (τ_{redeem} **≥ 7d**) during L2 panic sells.

3. Class A2 & B2: Wrapped Bitcoin (BTC-Anchored) & Tokenized RWAs (Gold & US Treasuries)

Pegged Asset	Real Anchor	Peg Mechanism & Custodian	Ethereum (L1) & Arbitrum (L2) Venues	Hedging & FISCO BCOS Role
WBTC, cbBTC, tBTC	**BTC** (1:1 Native Bitcoin)	Custodial (BitGo/Coinbase) & Threshold Bridge	**L1 & Arb:** UniV3 (cbBTC/WBTC), Curve, Camelot, Aave V3	Hedged via Short BTC-PERP or cross-arbitraged (cbBTC/WBTC basis).
PAXG & XAUT	**Gold** (1 Troy Oz XAU)	1:1 Physical London Good Delivery / Swiss Gold	**L1 & Arb:** UniV3 (PAXG/USDC, PAXG/XAUT), Curve	Hedged via XAU-PERP or delta-neutral PAXG/XAUT pair (τ_{redeem} **= T+2**).
USDY & OUSG (Ondo)	**US T-Bills** (Short-Term)	Yield-Accruing RWA Token (~4.5–5.0% APY)	**L1 & Arb:** Curve, Flux Finance, Arbitrum RWA Pools	Pledged as institutional RWA collateral inside FISCO BCOS Par-Capped Repo vaults.

System Architecture: Layer 1 Calibration

PegGuard is architected across five sequential layers that separate temporary liquidity mispricing from permanent capital impairment.

Layer 1: Dual-Anchor Fair-Value & Multi-Horizon Equilibrium Calibration

A robust valuation model must account for two realities simultaneously: (A) **Economic Fundamental Fair Value** grounded in reserve quality and redemption mechanics, and (B) **Empirical Market Consensus** capturing unobserved market-priced risks and persistent structural discounts.

1. Economic Fundamental Fair-Value Benchmark (FV_{fund})

Rather than assuming par value is always 1.000, the fundamental value $\mathbf{FV_{fund}(t)}$ discounts par value by verifiable reserve solvency, redemption rights, liquidity depth, and asset-specific cash flows (such as staking yield $\mathbf{y_{asset}}$ minus redemption fees $\mathbf{r_{fee}}$ over primary settlement delay $\boldsymbol{\tau_{redeem}}$):

$$FV_{fund}(t) = P_{par} \times \min(1, \text{LiquidReserves}_t / \text{CirculatingSupply}_t) \times \exp[\ (y_{asset} - r_{fee} - c_{illiquidity}) \times \tau_{redeem}\ ]$$

2. Multi-Horizon Market Consensus (μ_{mkt}) & Dual-Path Spot Triangulation

Even if an asset is fundamentally worth $1.00, the market may persistently price it at a discount (e.g., $0.985) due to regulatory friction. Moreover, during liquidity crunches, direct spot books (WBETH/ETH) frequently dislocate from synthetic cross-rates ((WBETH/USDT)/(ETH/USDT)). Layer 1 synthesizes a dual-path consensus ratio $\mathbf{R_{cons}(t)}$ and tracks empirical equilibrium across three rolling windows (1-Week $\boldsymbol{\mu_{1w}}$, 1-Month $\boldsymbol{\mu_{1m}}$, and 1-Year $\boldsymbol{\mu_{1y}}$):

$$R_{cons}(t) = w_{dir} \times R_{direct}(t) + (1 - w_{dir}) \times [\ P_{pegged/USDT}(t) / P_{anchor/USDT}(t)\ ]$$
$$\mu_{mkt}(t) = w_{1w} \times \text{Average}_{1\text{-week}}(t) + w_{1m} \times \text{Average}_{1\text{-month}}(t) + w_{1y} \times \text{Average}_{1\text{-year}}(t)$$

3. The Conservative Dual-Anchor Baseline (Baseline*(t))

Layer 1 synthesizes both anchors using a **Conservative Minimum Operator** to define the actionable convergence target **Baseline*(t)** and standardized dislocation **Z(t)** against 30-day rolling volatility $\boldsymbol{\sigma_{30D}(t)}$:

$$\text{Baseline}^*(t) = \min[\ FV_{fund}(t), \mu_{mkt}(t)\ ]$$
$$\text{Abnormal Spread } \delta(t) = R_{cons}(t) - \text{Baseline}^*(t), \qquad \text{Z-Score}(t) = \delta(t) / \sigma_{30D}(t)$$

Why the Dual-Anchor + Triangulation is Superior: (1) Dual-path triangulation detects cross-book liquidity basis anomalies instantly. (2) If the market persistently prices an asset at 0.98 while theoretical par is 1.00, **Baseline*(t) = 0.98**, avoiding dead trades. (3) Conversely, if moving averages sit at 1.00 but reserve impairment drops $\mathbf{FV_{fund}(t)}$ to 0.82 today, **Baseline*(t)** immediately drops to 0.82, blocking broken pegs.

3-Regime Duration & Signed Recovery-Risk EV Gate

Layer 2: 3-Regime Markov-Switching Duration & Fast OLS Fallback

Calibrating a single Ornstein-Uhlenbeck (O-U) reversion speed **θ** on historical data fails when a peg fractures, while pure Maximum-Likelihood Markov-Switching (Hamilton EM filter) can fail to converge on noisy or flat windows. Layer 2 uses a **Two-Stage Hybrid Estimator**: a deterministic **<15ms Closed-Form AR(1) OLS Engine** ($\Delta\delta_t = \alpha + \kappa\delta_{t-1} + \varepsilon_t$) with Delta-method standard errors that initializes and provides a real-time fallback for the **3-Regime Markov-Switching SDE** across latent states $S_t \in \{1, 2, 3\}$:

$$d\delta(t) = \theta(S_t) \times [\ \mu(S_t) - \delta(t)\ ]\ dt + \sigma(S_t) \times dW(t), \qquad t_{1/2}(S_t) = \ln(2) / \theta(S_t)$$

$$\text{Fast OLS Initializer \& Fallback: } \theta_{OLS} = -\ln(1 + \kappa) / \Delta t \qquad \text{(Death Spiral Canary: Veto } S_t = 3 \text{ if } \beta = 1 + \kappa \geq 1.0)$$

- **Regime 1 ($S_t = 1$, Normal Microstructure):** High reversion speed ($\theta_1 \gg 0$), low σ_1, rapid half-life (minutes to hours).
- **Regime 2 ($S_t = 2$, Temporary Liquidity Distress):** Reserves remain solvent ($FV_{fund} \approx 1$), but panic selling depletes order books (e.g., USDC during SVB or stETH during 3AC). Reversion θ_2 slows to days and σ_2 spikes.
- **Regime 3 ($S_t = 3$, Fundamental Peg Collapse):** Reserve insolvency or non-stationary divergence ($\beta \geq 1.0$, $\theta_3 \leq 0$, $t_{1/2} \to \infty$). Historical reversion parameters are invalidated and all buying is locked out.

Layer 3: Recovery-Risk EV & Signed Perpetual Funding Gate

Because PegGuard executes **Long Discounted Spot + Short Reference Perpetual**, perpetual funding is **directionally signed**: when the 8-hour funding rate $F_{8h} > 0$ (contango), longs pay shorts—meaning our short hedge **earns a positive funding rebate**! Carry only becomes a cost drag when $F_{8h} < 0$ (backwardation during panic shorting) or via cross-margin borrow interest (**BorrowCost**). Layer 3 weighs this signed net gain against **Probability of Recovery (P_{rec})**, **Loss Given Default (LGD)**, and non-linear panic slippage:

$$\text{SignedFundingCF} = +F_{8h} \times [\ T_{exp}(S_t) / 8h\ ] \qquad \text{(Yield Rebate if } F_{8h} > 0 \text{ on Short Perp; Cost Drag if } F_{8h} < 0)$$

$$\text{NetGain}_{rec} = |\delta(t)| + \text{SignedFundingCF} - \text{BorrowCost}(T_{exp}) - 2 \times \text{TakerFee} - \text{Slippage}(Q, \text{Depth}_t, \tau_{redeem})$$

$$\text{Recovery-Adjusted EV} = P_{rec} \times \text{NetGain}_{rec} - (1 - P_{rec}) \times \text{LGD}$$

The Execution Hurdle & Watermark Rule: A trade is authorized only if $S_t \neq 3$ ($\beta < 1.0$) and **Recovery-Adjusted EV ≥ HurdleRate (+15 bps)**. If EV falls below hurdle, Layer 3 updates $\texttt{WatermarkFloor}_i = R_{cons}(t) - \Delta$ and abstains until the price drops lower.

Layer 4: Stateful Asset-Specific Token Check

In peg arbitrage, broad crypto Fear & Greed and perpetual futures Open Interest (OI) are largely irrelevant because stablecoins and liquid staking tokens are shorted and arbitraged **on-chain and on spot order books**. Layer 4 executes a stateful **Asset-Specific Token Check ($T_t \in [0, 1]$)** over the delta window since the last inspection ($\Delta t = t_{now} - t_{last_check}$, defaulting to a **7-day lookback** on initialization):

1. Stateful FinBERT News & Regulatory Filing Sentinel (50% Weight + Hard Veto Gate)

Because fundamental solvency is the primary determinant of peg recovery, the dominant **50% weight** is assigned to our asynchronous NLP Sentinel. It evaluates all token-specific news, reserve attestations, custodian disclosures, and regulatory filings published within **Δt** using `ProsusAI/finbert` with Named Entity Recognition (NER) and exponential time-decay weighting ($w_i = \exp(-\lambda \Delta t_i)$):

- **Solvency vs. Liquidity Separation:** Distinguishes temporary banking or validator queue delays from permanent reserve impairment.
- **Binary Hard Veto ($I_{SafeVeto} = 0$):** If FinBERT detects structural insolvency, custodian bankruptcy, or a smart-contract exploit (**$P(\text{Insolvency}) \geq 0.60$**), it zeroes out the entire Token Check score immediately and locks `WatermarkFloor = 0.0`.

2. On-Chain & Spot Peg Microstructure Matrix (30% Weight)

Tracks the four exact mechanisms on Ethereum L1 and Arbitrum L2 where stablecoin and pegged-asset stress actually materializes:

- **Curve / DEX Pool Balance Tilt:** Monitors pool composition (e.g., Curve 3pool, stETH/ETH, or Arbitrum wstETH/ETH). Waits while pool skew is accelerating (>75% dumped asset) and signals entry when pool imbalance peaks and reverses.
- **DeFi Lending Utilization & Borrow APR Squeeze (Aave V3 / Morpho):** Whales short stablecoins by borrowing them on Aave. When utilization crosses the kink (>90%) and borrow APR spikes (>50% APR) while FinBERT confirms solvency, on-chain shorts are forced to buy back to repay debt.
- **Primary Mint / Burn Redemption Velocity:** Verifies that on-chain token burns are active during a discount, proving institutional arbitrageurs are successfully redeeming at $1.00 par.
- **Spot Order-Book Imbalance (OFI) & Bid-Wall Refill:** Confirms bid-to-ask depth recovery within 0.5% of mid-price.

3. Grok API Dual-Persona Token Check (20% Weight)

Queries the Grok API on live X/Twitter discourse for the target asset: acts **contrarian** on emotional retail panic, but flips to a **strict directional veto** if security monitors (PeckShield/Lookonchain) flag an active exploit or issuer freeze.

$$\text{TokenScore } T_t = I_{SafeVeto} \times [\, 0.50 \times F_{FinBERT} + 0.30 \times M_{OnChain+Spot} + 0.20 \times G_{Grok} \,], \quad \text{where } I_{SafeVeto} = 0 \text{ if } P(\text{Insolvency} \mid \text{FinBERT}) \geq 0.60$$

Layer 5: Asset-Specific Hedging & Contagion Control

A single naive delta-neutral mechanism cannot be applied blindly across all pegged assets. Shorting BTC or ETH against a USD stablecoin introduces severe directional market risk, whereas shorting USDT on a retail perpetual exchange exposes the arbitrageur to flight-to-safety short squeezes (when panic buyers bid USDT to $1.04) and exchange default. Layer 5 enforces **Asset-Specific Hedging**:

1. Asset-Specific Execution & The FISCO BCOS Par-Capped USDT Repo

- **Class A — Crypto-Pegged Derivatives (e.g., stETH/ETH, wstETH/ETH, WBETH/ETH, WBTC/BTC):** Executes simultaneous Long Spot on Arbitrum/Ethereum/CEX + Short Reference Perpetual (**ETH-PERP** or **BTC-PERP**), neutralizing underlying crypto delta while locking in the structural staking or custody basis.
- **Class B — Fiat-Pegged Stablecoins via FISCO BCOS Institutional Par-Capped Repo (e.g., USDC/USDT, DAI/USDC):** To hedge a discounted long position (e.g., buying **USDC** at $0.90) without directional crypto exposure or retail order-book squeeze risk, PegGuard routes the short leg through an **On-Chain Institutional Repo & Settlement Facility built on FISCO BCOS / EVM smart contracts**:
  • *Par-Capped Liability (Zero Short-Squeeze Risk):* The borrowed **USDT** debt obligation is programmatically anchored and capped at its legal primary redemption par value ($1.000). Even if retail exchange order books temporarily spike **USDT** to $1.04 during a flight to safety, the smart-contract liability cannot exceed 1.000, eliminating artificial margin liquidations.
  • *Asymmetric Downside Convexity:* If the shorted stablecoin (**USDT**) suffers its own de-peg below $1.00, the repayment liability shrinks proportionally, turning counterparty de-peg risk into an asymmetric hedge gain inside an isolated non-custodial vault.

> $$\text{Effective Short Liability}(t) = \min[\, P_{USDT}(t),\ 1.000\,] \times Q_{borrowed}$$ (Eliminates >$1.00 Retail Squeeze & Captures Downside De-Peg Gain)

2. Cross-Stablecoin Contagion Matrix & Convex Position Sizing

De-pegging events are rarely isolated: when USDC de-pegged in March 2023, DAI and FRAX simultaneously de-pegged due to shared reserve collateral and Curve 3pool liquidity coupling. PegGuard maintains a dynamic **Contagion Exposure Matrix (β_{ij})** measuring reserve overlap and pool co-dependency across stablecoins **i, j**:

> $$\text{OrderSize}_i = \text{BaseSize} \times [\, 1 + \alpha \times (|Z_i| - \text{Threshold})^2\,] \times T_{i,t} \times \max[\, 0,\ 1 - \textstyle\sum_{j \neq i} \beta_{ij} \times \text{DistressProb}_j\,]$$

> **Portfolio Contagion Brake:** If Stablecoin **j** enters Regime 2 or 3 distress, the term $\sum \beta_{ij}\ \text{DistressProb}_j$ automatically scales down capital allocation across all correlated stablecoins **i**, protecting the portfolio from systemic domino cascades.

Comparative Analysis

The table below benchmarks the upgraded **PegGuard** architecture against conventional retail bots and traditional academic statistical arbitrage models:

Dimension	Standard Retail / Grid Bot	Conventional Stat-Arb (Moving Average + O-U)	Upgraded PegGuard Framework
Scanning & State Memory	Stateless polling; spams orders on static price grids.	Re-runs full model on every tick regardless of prior rejection.	**Autonomous Master Toggle + Stateful Watermark Floor** (re-evaluates only when price drops lower).
Fair-Value Benchmark	Rigid static par ($1.000).	Pure historical moving averages (blind to reserve collapse).	**Conservative Dual-Anchor** $\min(FV_{fund}, \mu_{mkt})$ + **Dual-Path Spot Triangulation** (R_{cons}).
Regime & Duration Modeling	None.	Single-state O-U process assuming constant reversion speed θ.	**3-Regime Markov-Switching SDE** with deterministic **<15ms Closed-Form AR(1) OLS Fallback**.
EV & Microstructure Gate	Ignores funding, depth, and default risk.	Adjusts only for static transaction fees; treats all funding as cost.	**Recovery-Risk EV** integrating P_{rec}, **LGD**, **Signed Perp Funding** (rebate vs. drag), and depth decay.
Signal & Adverse Selection	Price-only triggers; buys falling knives to $0.00.	Lagging price stop-losses after capital impairment.	**Asset-Specific Token Check:** FinBERT Sentinel (50% + Veto) + Curve/Aave/Burn Flow (30%) + Grok (20%).
Hedging & Contagion	Unhedged directional exposure.	Naive perp shorting (vulnerable to >$1.00 squeezes & contagion).	**Asset-Specific Hedge:** FISCO BCOS Par-Capped USDT Repo + Cross-Coin Contagion Matrix (β_{ij}).

Core Scientific & Financial Contributions

1. **Autonomous Watermark-Guided Scanning:** Eliminates redundant econometric/NLP re-computation on shallow de-pegs by caching the rejected price floor ($\texttt{WatermarkFloor}_i$) and waking Layers 2–4 only when the discount deepens.
2. **Distinguishing Liquidity Distress from Capital Impairment:** Replaces naive mean-reversion assumptions with a hybrid OLS + 3-Regime Markov framework (P_{rec} vs. **LGD**) and directionally signed perpetual funding cashflows.
3. **Dual-Anchor Valuation & Par-Capped Consortium Repo:** Prevents false-positive entries on discounted/insolvent assets while solving the stablecoin hedging paradox via FISCO BCOS par-bounded debt settlement and Contagion Matrix (β_{ij}) throttling.

Empirical Evaluation & Stress-Testing Plan

The framework is evaluated using tick-level spot order-book replay across Ethereum, Arbitrum, and Binance, on-chain DeFi telemetry (Curve pool balances, Aave borrow rates, primary mint/burn events), and timestamped news archives across four crisis episodes, comparing **Model A (Naive Stat-Arb)**, **Model B (Single-Regime O-U + Fees)**, and **Model C (Upgraded PegGuard)**:

Historical Crisis	Asset Scope	Market & Microstructure Dynamics	PegGuard Validation Outcome
SVB Collapse & Multi-Coin Contagion (March 2023)	`USDC, DAI, FRAX / USDT`	USDC dropped to $0.88; Curve 3pool tilted >80% USDC/DAI; retail USDT spiked to $1.04.	**Regime 2 + Contagion Proof:** Watermark skips shallow $0.994 dip; triggers at deeper discount. Contagion matrix (β_{ij}) halts double-dipping in DAI/FRAX; FISCO BCOS par-capped hedge prevents $1.04 USDT squeeze.
Terra / UST Terminal Collapse (May 2022)	`UST / USD`	Mint-burn death spiral; FV_{fund} collapsed to $0.00 while 30-day moving average remained near $1.00.	**Regime 3 Survival Proof:** Models A & B suffer -100% loss relying on historical θ. PegGuard switches to Regime 3 ($I_{SafeVeto} = 0$, `WatermarkFloor = 0.0`), preserving 100% of capital.
stETH & wstETH Liquidity Crisis (June 2022)	`stETH, wstETH / ETH`	7% discount from 3AC forced liquidations & Curve stETH pool skew; Beacon ETH reserves remained 100% solvent.	**FinBERT + On-Chain Flow Proof:** FinBERT (50% weight) confirms zero protocol insolvency; Curve pool tilt stabilization signals exhaustion; ETH-PERP hedge harvests spread safely.
Order-Book Depth & Panic Slippage Stress Test	All Target Pairs ($10k–$10M)	Simulates order-book depth contraction (80% liquidity drop) and T+3 redemption bottlenecks during market panic.	**Microstructure Decay Proof:** Quantifies the exact capital capacity threshold where panic slippage and redemption lag erase theoretical arbitrage alpha.

Core Scientific Evaluation Metrics

- **Recovery vs. Impairment Classification Accuracy:** Brier score and ROC-AUC of the 3-Regime classifier (P_{rec}) in separating temporary distress (SVB, 3AC) from terminal collapse (UST).
- **Watermark Computational Efficiency Gain:** Percentage reduction in redundant Layer 2–4 invocations and API calls achieved by Stateful Unprofitable-Price Watermarking during multi-day de-peg events.
- **Capacity, Slippage & Contagion-Adjusted Net Sharpe:** Risk-adjusted return and Maximum Drawdown (MDD) reduction after accounting for L2 depth, signed funding carry, and the Cross-Stablecoin Contagion Matrix (β_{ij}).

Project Roadmap & References

Sequential Implementation Roadmap

1. **Phase 1 (Autonomous Watchlist Scanner & Microstructure Ingestion):** Build the user-toggled daemon loop, Stateful Unprofitable-Price Watermark Cache ($\texttt{WatermarkFloor}_i$), dual-path spot klines, L2 order-book depth & OFI (`ccxt`), Curve pool tilts, Aave borrow utilization rates, and primary mint/burn telemetry across Ethereum and Arbitrum.
2. **Phase 2 (Dual-Anchor, OLS-Initialized 3-Regime & Signed Carry Core):** Implement Fundamental Fair-Value ($\mathbf{FV_{fund}}$), Dual-Path Multi-Horizon Consensus ($\boldsymbol{\mu_{mkt}}$), the <15ms AR(1) OLS initializer + 3-Regime Markov-Switching O-U engine, and the Signed Perpetual Funding Recovery-Risk EV gate.
3. **Phase 3 (Multi-Modal NLP & Grok Intelligence Layer):** Deploy the stateful delta-window scanner ($\mathbf{t_{last_check}}$ / 7-day lookback) combining time-decayed `ProsusAI/finbert` regulatory/news inference with structured Grok API sentiment decomposition.
4. **Phase 4 (FISCO BCOS Par-Repo & Contagion Engine):** Implement the asset-specific execution router (Ethereum/Arbitrum + CEX), FISCO BCOS / EVM par-capped debt settlement module, and Cross-Stablecoin Contagion Matrix ($\boldsymbol{\beta_{ij}}$).
5. **Phase 5 (Panic Depth Stress-Testing & Manuscript):** Run tick-level historical replays across SVB, UST, 3AC, and WBETH crises, quantifying watermark efficiency and arbitrage capacity decay under order-book illiquidity.

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